Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/31/2008
PI Name:	Mark, Roger G Ph.D.		
Project Title:	Computational Models of Cardiovascular Function for Simulation, Data Integration, and Clinical Decision Support		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rgmark@mit.edu	**Fax:**	FY 617-258-7859
PI Organization Type:	UNIVERSITY	**Phone:**	617-253-7818
Organization Name:	Massachusetts Institute of Technology		
PI Address 1:	Room E25-505		
PI Address 2:	Massachusetts Institute of Technology		
PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02139	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	06/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Heldt, Thomas (Massachusetts Institute of Technology) Verghese, George (Massachusetts Institute of Technology)		
Grant/Contract No.:	NCC 9-58-CA00403		
Performance Goal No.:			
Performance Goal Text:			
	One of the highest priority problems in the current manned space program is orthostatic intolerance (OI) experienced by astronauts upon their return to the normal gravitational environment. This problem has been well known since the earliest days of manned space flight, and has been intensely studied in-flight and in many ground-based (bed-rest) studies. A number of countermeasures have been proposed and evaluated, but to date no effective and practical countermeasure has been developed. While the effectiveness of countermeasures can only be explored in ground-based experiments and ultimately be proven successful when used by astronauts, the small number of subjects studied in both situations makes the interpretation of limited experimental data for countermeasure development and refinement difficult. Mathematical models that represent the essential functional components of the cardiovascular system provide a		

Task Description:

powerful means to extract physiologically relevant information from multiparameter experimental data.

We will extend progress already made, with the following specific aims:

1. Develop procedures and algorithms for automated model-based extraction of physiologically relevant information from multivariate data streams, particularly those relevant to countermeasure development for OI. We will expand our unified data archive of hemodynamic signals during gravitational stress.

2. Develop strategies for systematic and effective model identification, i.e., for the adaptation of model structures and parameters to the characteristics of the available physiological data. These strategies will be based on analytical and computational studies of candidate models and their sensitivities to parameter changes.

3. Apply our computational modeling and parameter identification technology to the challenge of cardiovascular monitoring of critically ill patients. Hemodynamic measurements will be transformed into physiologically meaningful parameters through model-based processing.

4. Develop and thoroughly test a graphical user interface for our current cardiovascular model. The resultant cardiovascular simulator will be open-source and capable of running on all major operating systems. Furthermore, the entire model will be made publicly available to researchers and educators.

The key findings and deliverables of the project are as follows:

1. Post-spaceflight orthostatic intolerance:

Our simulations indicate that post-spaceflight orthostatic intolerance can be explained either on the basis of a sufficiently severe degree of hypovolemia or by a moderate (and by itself benign) degree of hypovolemia coupled to alterations in 'secondary' cardiovascular parameters. Our simulations further indicate that alterations in the vasoconstriction loop of the arterial baroreflex can lead to a particularly dramatic collapse of arterial blood pressure homeostasis in the upright posture. These results are corroborated by ground-based studies on bedrest subjects and in-flight experiments on astronauts that suggest that midodrine (an alpha1-adrenergic agonist acting to enhance peripheral vasoconstriction) is effective in preventing orthostatic intolerance in otherwise susceptible individuals. Our results suggest a countermeasure strategy aimed at two targets: the underlying hypovolemia or the secondary mechanisms. Midodrine affects the secondary mechanism of arterial baroreflex dysfunction.

2. Model-based patient monitoring:

We have developed estimation algorithms that have the potential to improve significantly the way in which hemodynamic variables are being monitored in data-rich clinical environments such as the intensive care unit. Using a single, continuously available arterial blood pressure waveform signal, we estimate continuously cardiac output, left ventricular ejection fraction, and left ventricular end-diastolic volume. These algorithms have been tested against both animal (canine and porcine hemorrhage models) and human data (taken from intensive care patients). Two reference measurements for each algorithm allow for absolute tracking of the estimated variables. While primarily intended for use in ground-based clinical environments, these algorithms can also be used to track astronaut hemodynamic performance while in space or on the lunar surface.

3. Platform-independent, user friendly simulator:

We have developed and tested a platform-independent, user friendly version of both a teaching version and a research version of our main cardiovascular model. The teaching version has been tested to great reviews with students at one of MIT's quantitative physiology courses. It has been made available to the general population through the PhysioNet internet portal and can be accessed at http://physionet.org/ . This site has already been visited over 1,200 times over the past five months.

The results obtained during the current funding cycle demonstrate impressively the utility of the models in integrating and interpreting hemodynamic data from astronauts and intensive care patients alike.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Outside the scope of modeling the effects of microgravity, numerical cardiovascular simulations:
1. provide a basis for interpretation of clinical data in the context of intensive care. Simulations are basic components of an intelligent patient monitoring system which can couple real-time physiologic data to cardiovascular pathology.

2. provide a powerful environment for teaching of normal and abnormal cardiovascular physiology.

This Final Report summarizes and thus concludes the third round of funding for this project from the National Space Biomedical Research Institute. The premise of our research is that computational models of the cardiovascular system are powerful adjuncts to ground-based and in-flight experiments. They provide a rational framework that quantitatively defines interactions between cardiovascular parameters and variables, supports the critical interpretation of limited experimental and clinical information, enables the testing of scientific hypotheses especially in a setting where experimental data come at a very high cost, and permits prediction of the impact of a specific countermeasure.
The third funding cycle was dedicated to an in-depth testing of hypotheses of the mechanisms underlying post-spaceflight orthostatic intolerance, the evaluation of mathematical methods to estimate cardiovascular parameters from individual astronauts' heart rate and blood pressure recordings, the adaptation of cardiovascular models for patient monitoring and clinical decision support in intensive care and spaceflight environments, and the development of a computational platform-independent, user-friendly version of the simulators generated under this research contract.

Post-spaceflight orthostatic intolerance:

Post-spaceflight orthostatic intolerance (OI) remains one of the highest-priority unsolved cardiovascular problems in the current manned spaceflight program as evidenced by a sizable number of scientific hypotheses still under investigation to elucidate the mechanisms underlying this syndrome. We have used our models to investigate systematically the effect alterations in particular cardiovascular parameters have on the cardiovascular system's ability to

maintain blood pressure homeostasis in the upright posture. We considered hypovolemia, increased compliance of the leg veins, cardiac atrophy, and alterations in the neural control of arteries and veins - all of which have been suggested as possible etiologies of post-spaceflight OI in the medical literature. Our simulations confirm that a sufficient degree of hypovolemia (~ 10% loss of total blood volume) by itself impairs blood pressure homeostasis to the degree seen in pre-syncopal astronauts on landing day, namely a drop in mean arterial blood pressure of up to 30 mm Hg five minutes after assuming the upright posture. Unrealistic degrees of changes in cardiovascular parameters representing the other hypotheses have to be invoked in order to achieve a similar detrimental effect on blood pressure homeostasis. If, however, such parameter changes are superimposed on a moderate degree of hypovolemia, the changes in secondary parameters become very important and can lead to a dramatic collapse of blood pressure homeostasis even in situations in which hypovolemia by itself is still tolerated well by the cardiovascular system. Of particular note is the combination of a moderate degree of hypovolemia and a reduction in the cardiovascular neural reflex system's ability to constrict arteries and veins in response to the upright posture: in the light of an otherwise benign level of hypovolemia, a 15% reduction in the reflex system's ability to vasoconstrict (i.e. a 15% reduction in the respective gain values) causes blood pressure to drop precipitously in the upright posture. These results point to a possible two-step mechanism of post-spaceflight orthostatic intolerance: a sufficient reduction in total blood volume by itself can cause post-spaceflight OI or a combination of a moderate degree of hypovolemia coupled with detrimental changes in secondary parameters can lead to the same syndrome. The implication of the latter insight is that effective countermeasures against OI can target the underlying hypovolemia or the secondary mechanisms.

Our suggestion that the arterial baroreflex-to-vasoconstriction loop is a main contributor to post-spaceflight OI has been corroborated by the identification of midodrine (an alpha1-adrenergic agonist) as a potential countermeasure against OI in bedrest studies (Ramsdell JAP, 2001) and flight studies (Platts et al. Aviat. Space Environ. Med. 2004, Platts et al. Aviat. Space Environ. Med. 2006).

Task Progress:

Model identification:

In the context of spaceflight-induced changes in cardiovascular performance, we sought to identify astronaut-specific cardiovascular parameters before spaceflight and on landing day to derive inference -- on a subject-by-subject basis -- on which cardiovascular parameters change significantly in response to exposure to microgravity. This task is mathematically difficult as experimental data commonly do not contain enough information to identify all the parameters of a given mathematical model. Identifying all parameters is not prudent in such cases as the resulting parameter estimates become very sensitive to minute variations (noise) in the data. To overcome this problem, we have refined, implemented, and evaluated a subset selection algorithm that allows for identification of a subset of parameters that can be estimated reliably from available data streams. When evaluating the algorithm's performance on simulated, noise corrupted data generated by our models, we achieved a maximum relative estimation error of (0.9 +/- 0.3)% if the estimation was constrained over the four most important parameters (out of a total of 140) as identified by the algorithm. In comparison, when we attempted to estimate fifteen parameters, the maximum relative estimation error rose to over 50%.

When we applied the estimation algorithm to recordings of mean arterial pressure and heart rate in pre- and post-spaceflight stand test data, the algorithm identified major changes in total blood volume and parameters associated with the baseline operation of the arterial baroreflex.

Model-based patient monitoring:

Computational models of the cardiovascular system can play an important role in integrating and interpreting hemodynamic data streams in data-rich environments such as in emergency care, peri-operative care, and intensive care. We have adapted the structure of the models developed under the previous and current funding cycle to allow for identification of important hemodynamic variables from continuous measurements of arterial blood pressure. We tested the resultant estimation algorithms against animal data (canine and porcine hemorrhage models) and against an extensive database of human intensive care patients (the Multiparameter Intelligent Monitoring in Intensive Care II Database). Using at most two reference measurements for calibration, we are able to track cardiac output, ejection fraction, and left ventricular end-diastolic volume with root mean square normalized errors of 14.8% (human data), 12.6% (dog data), and 11.9% (dog data), respectively. The main advantage of our approach is the availability of cardiac output, ejection fraction, and left ventricular end-diastolic volume estimates on a continuous basis from minimally- to non-invasively recorded arterial blood pressure waveforms. Two disclosures and two provisional patents have been filed on the estimation algorithms.

Platform-independent, user friendly simulator:

We have developed and tested a platform-independent, user friendly version of both a teaching version and a research version of our main cardiovascular model. The teaching version has been tested to great reviews with students at one of MIT's quantitative physiology courses. It has been made available to the general population through the PhysioNet internet portal and can be accessed at http://physionet.org/ . This site has already been visited over 1,200 times over the past five months.

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Patents	TBD. Patent TBD, June 2007. Jun-2007 Parlikar TA, Ranade GV, Heldt T, Verghese GC. "Model-based method for continuous determination of cardiac output and Peripheral Vascular Resistance."